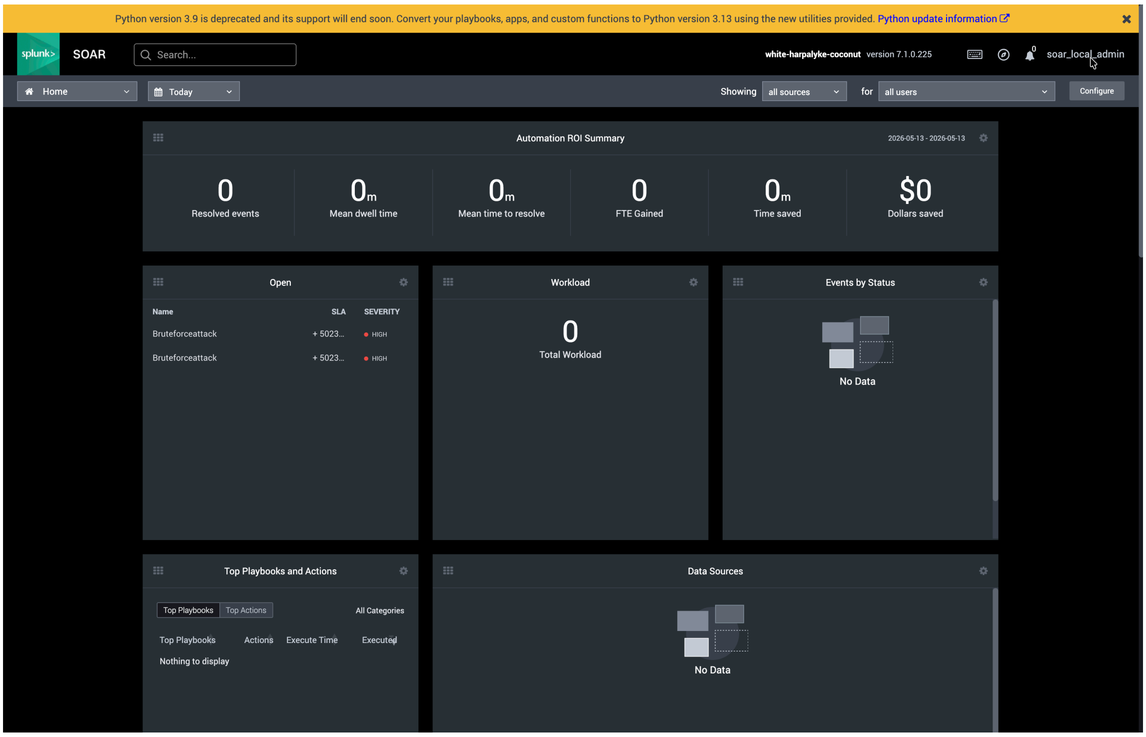
Creating a Privilege Escalation Playbook in SOAR

Summary

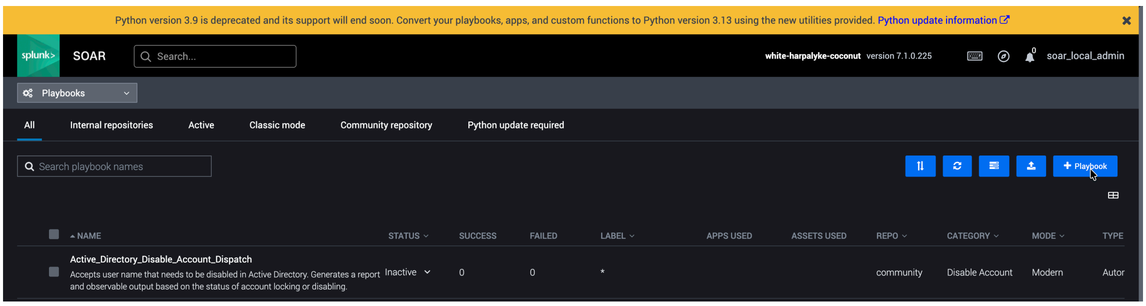
This document details the steps involved in Creating and Configuring the Privilege Escalation Use Case Playbook with all the required blocks added and configured and executed to validate the SOAR playbook by simulating the Privilege Escalation attack.

1. Login to Splunk SOAR as Admin

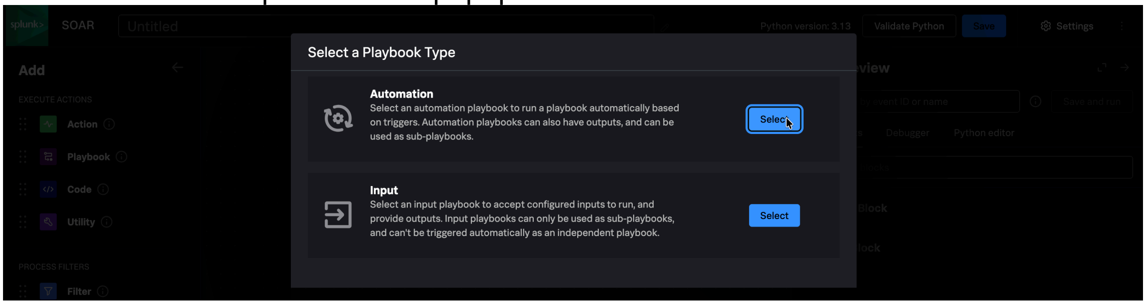


2. Create a new Playbook

1) Go to `Home > Playbooks > + Playbook`

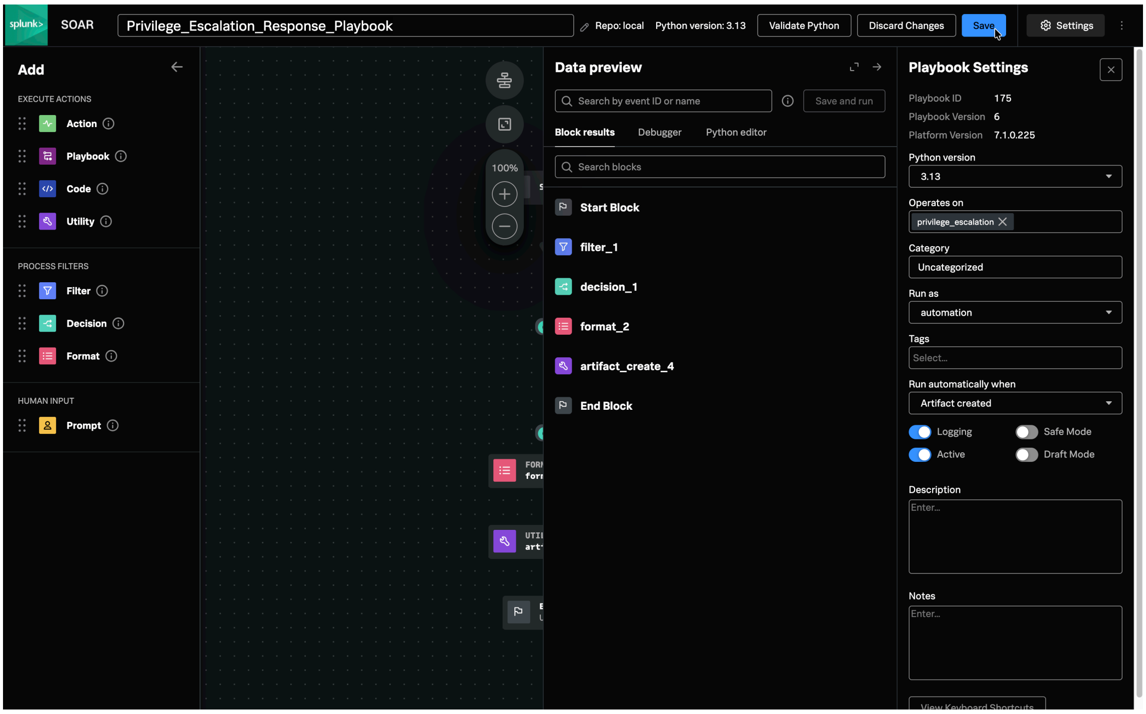


2) Select Automation in Playbook Type popup
A **new tab** will open with this popup. Select **Automation**

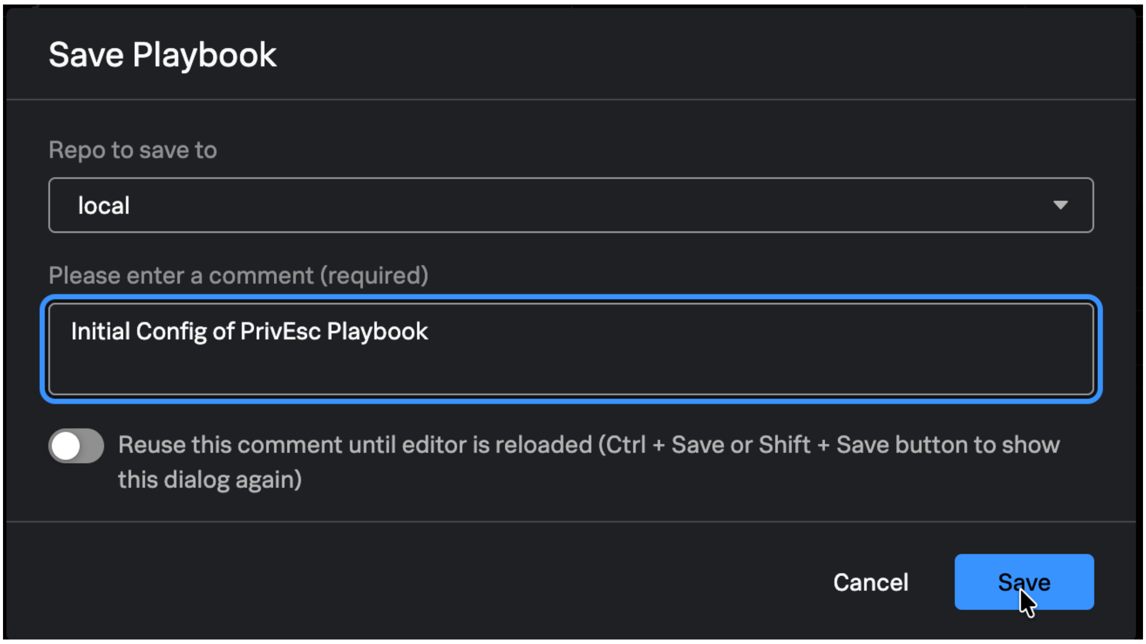


3) Configure Playbook Settings
Click **Settings**,
Do the following Configuration and Save

```
Name: Privilege_Escalation_Response_Playbook
Operates on: privilege_escalation from drop-down
Run as: automation from drop-down
Run automatically when: Artifact created
Enable => Logging, Active
```

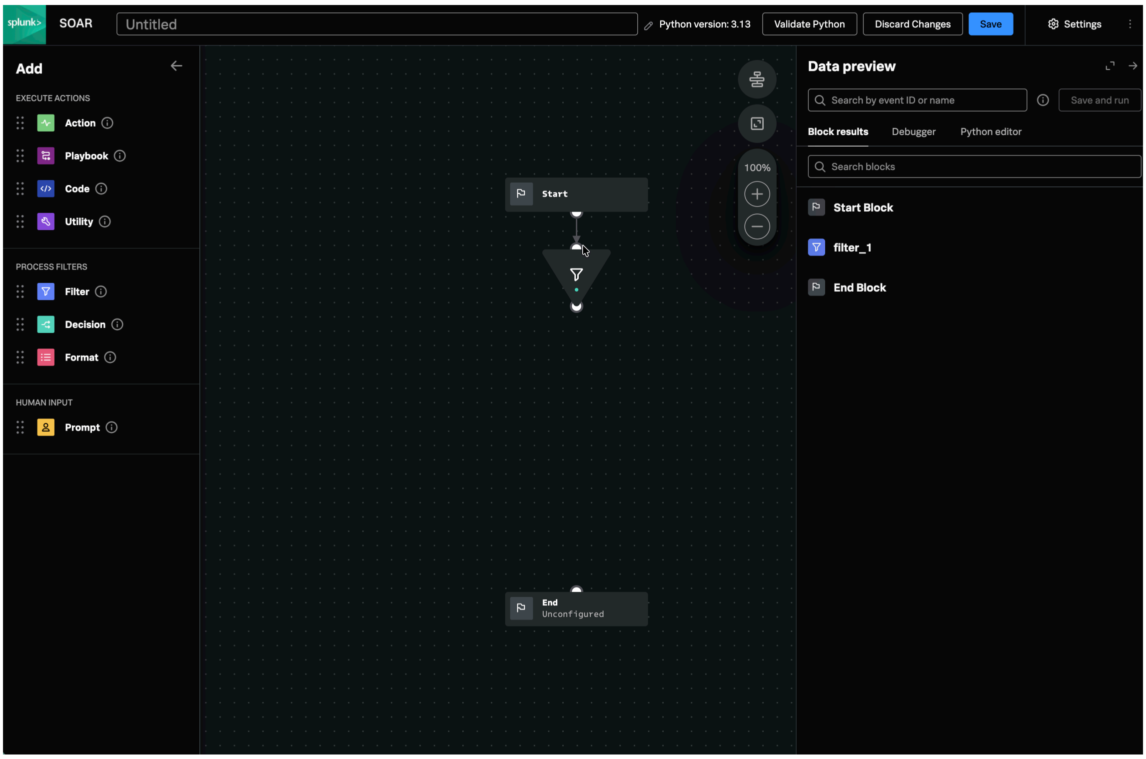


4) **Save the Initial Config**
Click **Save**
Repo to save to: local
Please enter a Comment (required): Initial Config of PrivEsc Playbook



3. Add Filter block and Configure

1) Click **x** to close the Playbook Settings panel on the right.
2) Drag and Drop **Filter block** on the playbook between **Start and End blocks** and **Connect it** with the Start block.



3) **Configure the Filter block**

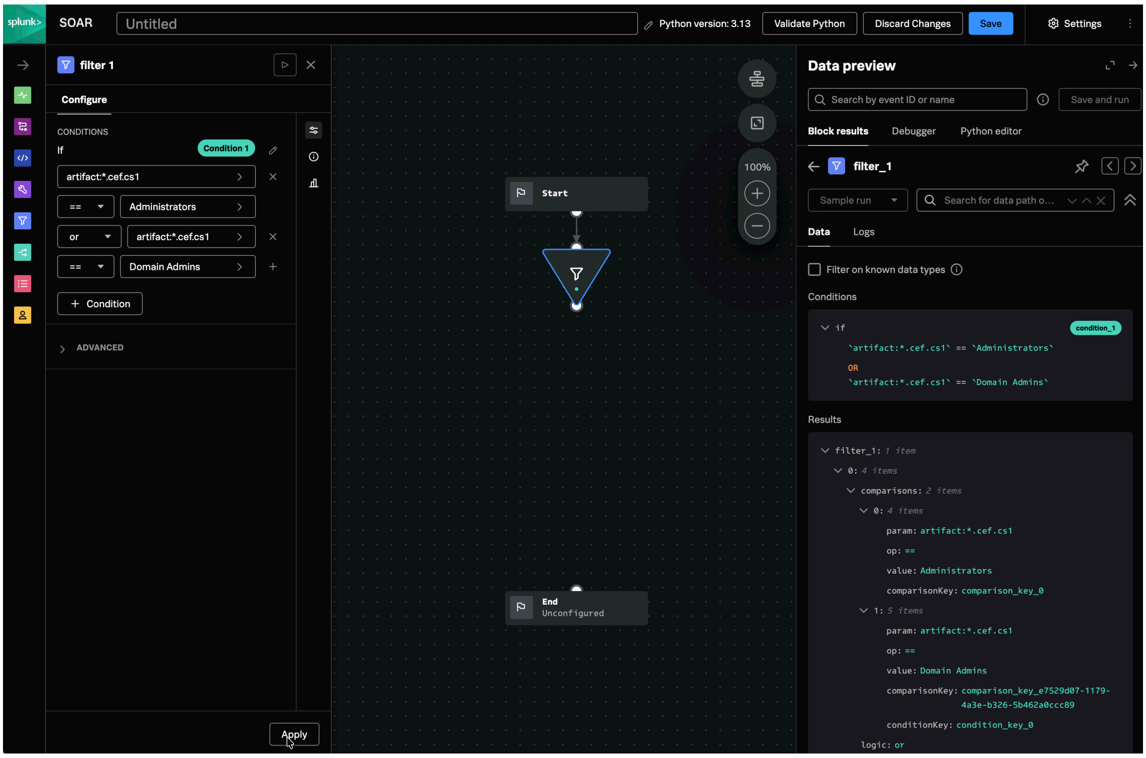
i) Enter the Condition to execute

Here **cs1** is the **group_name** value sent from the **Event Forwarding**.
This block filters out the events based on the `group_name` added.

Configure CONDITIONS

```
If
artifact:*.cef.cs1 == Administrators
or
artifact:*.cef.cs1 == Domain Admins
```

Click **Apply**



ii) **Save the changes to the Playbook**

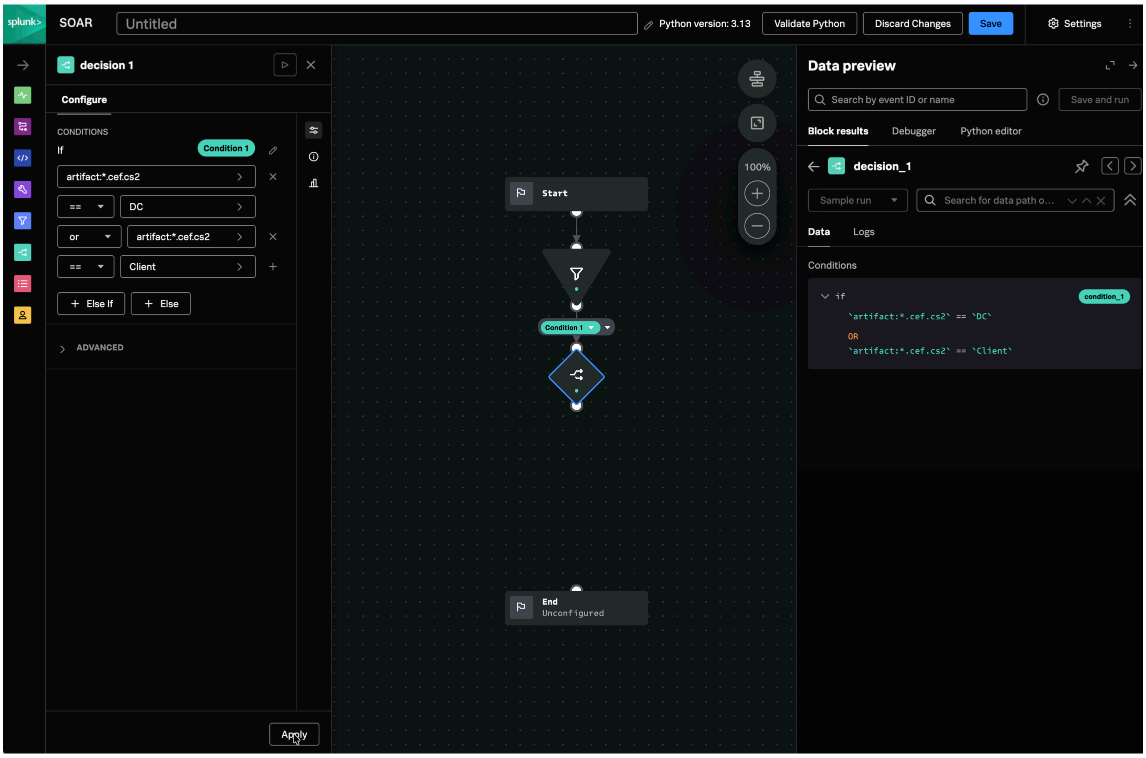
Click **Save**

Repo to save to: local
Please enter a Comment (required): Added and Configured Filter block in the playbook > Save

4. Add Decision block and Configure

1) Drag and drop the **Decision block** below the **Filter block** and **Connect** them.
2) **Configure the Decision block**
   i) Enter the Condition to execute
   Here **cs2** is the **source** value sent from the **Event Forwarding.**
   This block decides if the event was performed on the source, **DC or Client**

   Configure CONDITIONS
   ```
   If
   artifact:*.cef.cs2 == DC
   or
   artifact:*.cef.cs2 == Client
   ```
   Click **Apply**



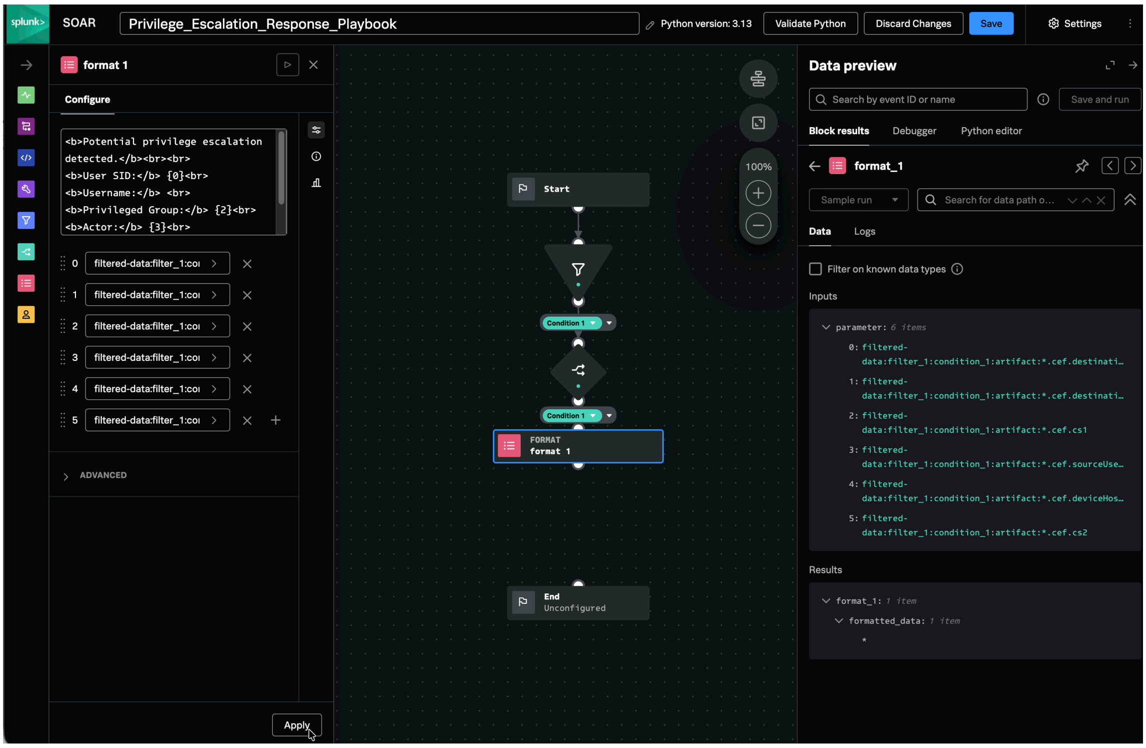
   ii) Save the changes to the Playbook
   Click **Save**
   Repo to save to: local
   Please enter a Comment (required): Added and Configured Decision block in the PrivEsc Playbook. > Save

5. Add Format block and Configure

1) Drag and drop the **Format block** below the **Decision block** and **connect** them.
2) **Configure the Format block**
   Format block organizes the data in a specific format and is configured as Input. It is later sent as output to the other blocks that are connected to it.

   i) **Enter the following**
      Configure:
```
<b>Potential privilege escalation
detected.</b><br><br>
<b>User SID:</b> {0}<br>
<b>Username:</b> {1}<br>
<b>Privileged Group:</b> {2}<br>
<b>Actor:</b> {3}<br>
<b>Host:</b> {4}<br>
<b>Source:</b> {5}
```

      Parameters:
```
0: filtered-
data:filter_1:condition_1:artifact:*.cef.destinationU
serId
1: filtered-
data:filter_1:condition_1:artifact:*.cef.destinationU
serName
2: filtered-
data:filter_1:condition_1:artifact:*.cef.cs1
3: filtered-
data:filter_1:condition_1:artifact:*.cef.sourceUserNa
me
4: filtered-
data:filter_1:condition_1:artifact:*.cef.deviceHostna
me
5: filtered-
data:filter_1:condition_1:artifact:*.cef.cs2
```
      Click **Apply**



ii) Save the changes to the Playbook

Click **Save**

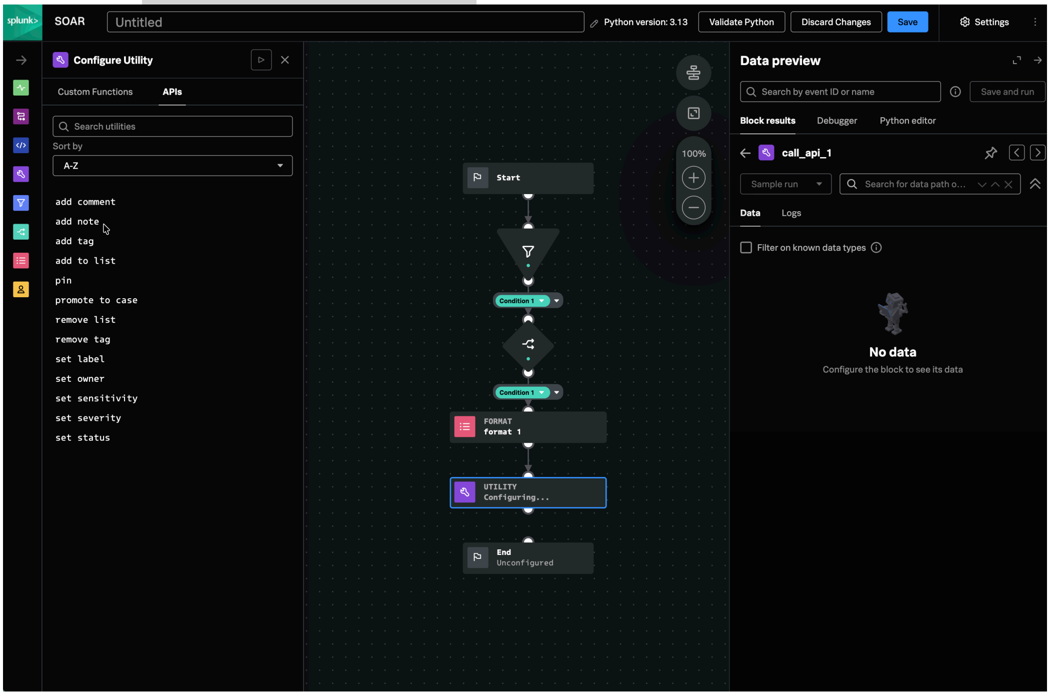
Repo to save to: local
Please enter a Comment (required): Added and Configured Format block in the PrivEsc Playbook. > Save

6. Add Utility blocks and Configure them

1) Add add note Utility block

i) Drag and drop the **Utility block** below the **Format block** and **connect** them.

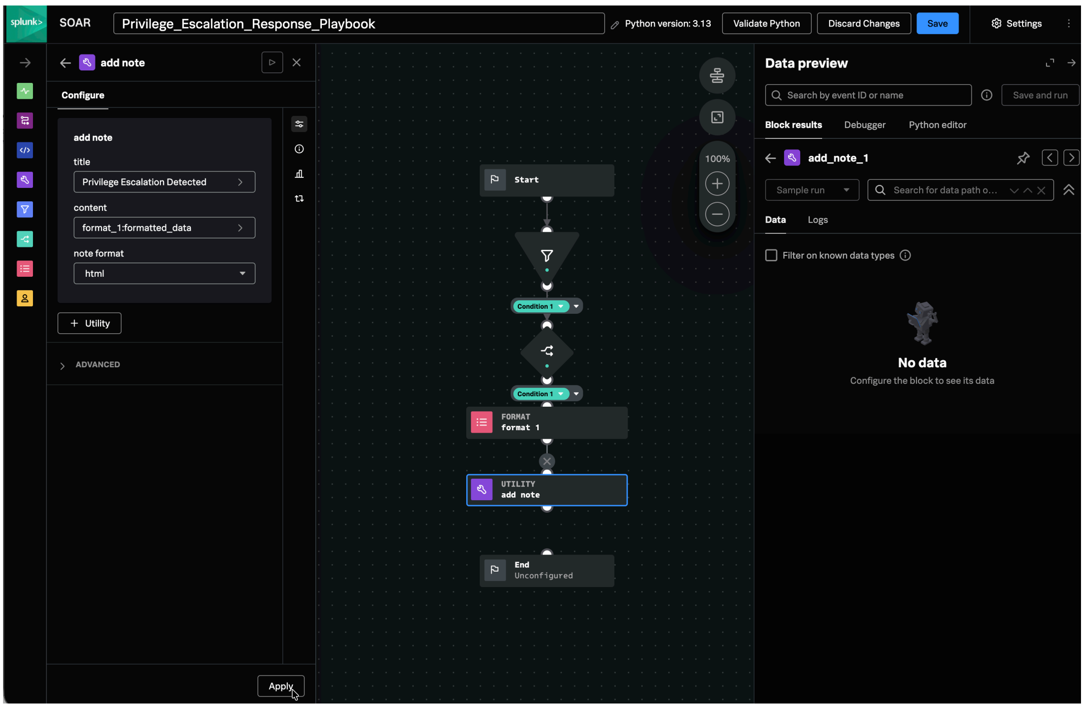
ii) **Configure the Utility block**
Go to `APIs tab > add note`



iii) **Configure add note**

```
title: Privilege Escalation Detected
content: format_1:formatted_data (select from the drop-down)
note format: html
```

Click **Apply**

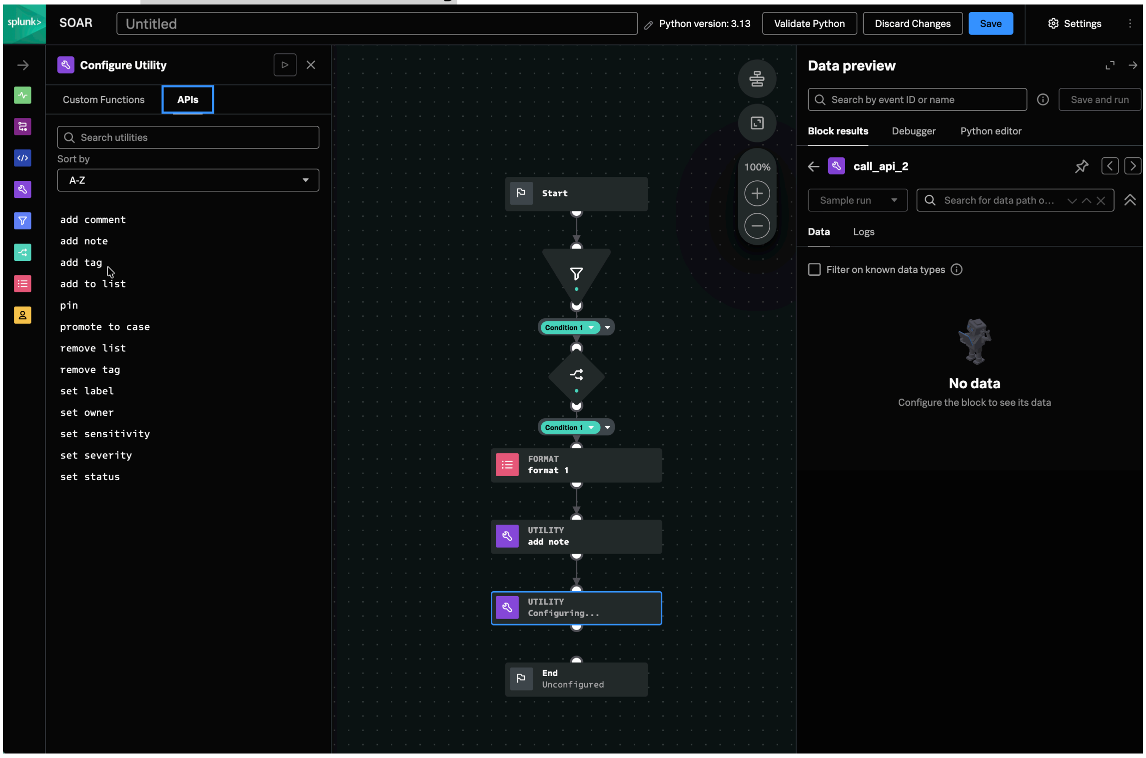


2) Add add tag Utility block

i) Drag and drop **Utility block** below the **add note Utility block** and **connect** them.

ii) **Configure the Utility block**

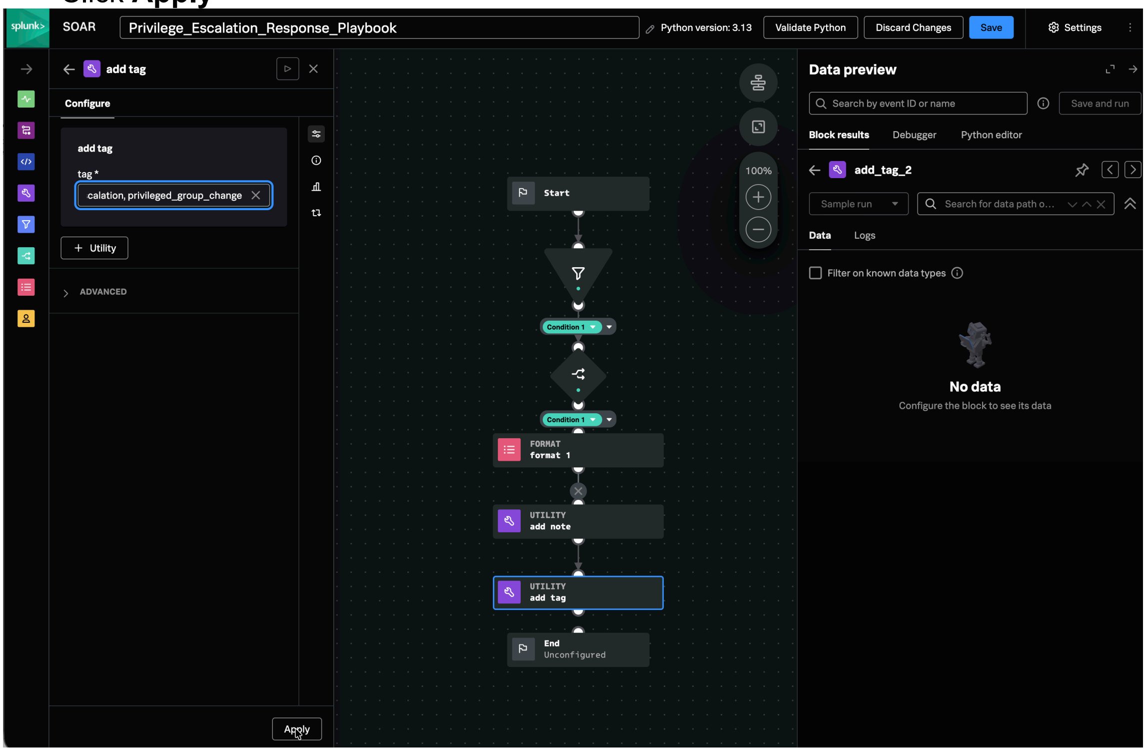
Go to `APIs tab > add tag`



iii) Configure add tag

`tag:` `privilege_escalation, privileged_group_change`

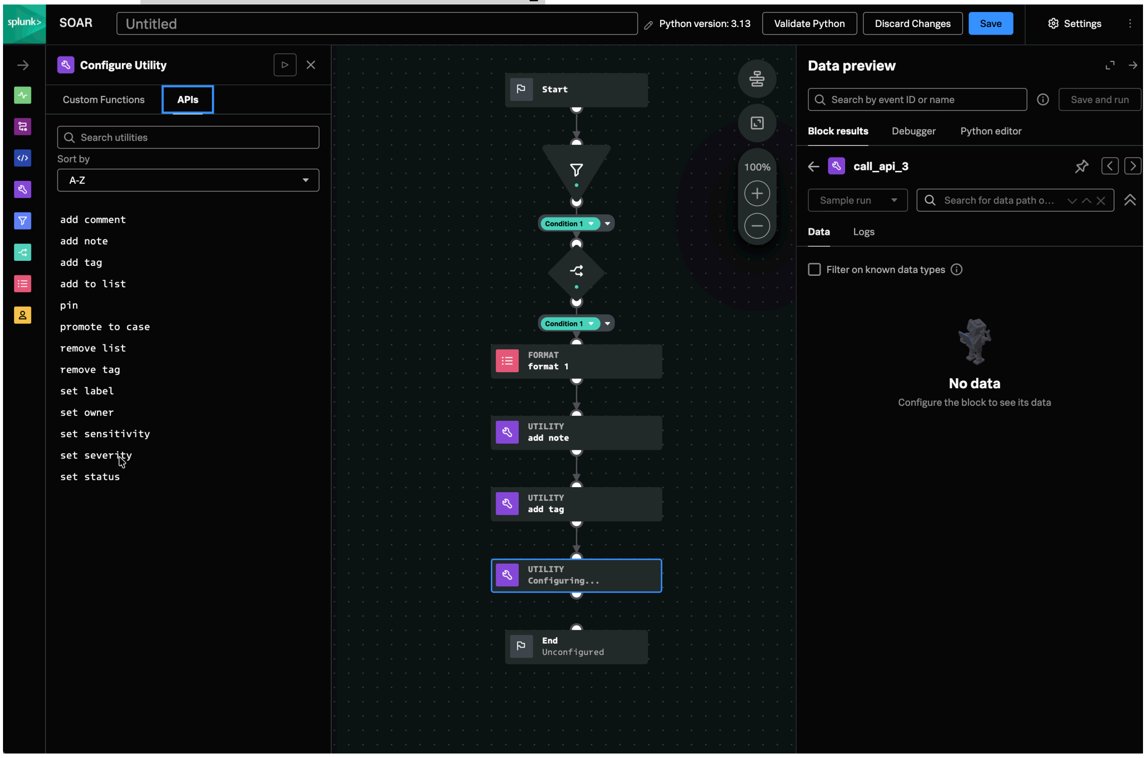
Click **Apply**



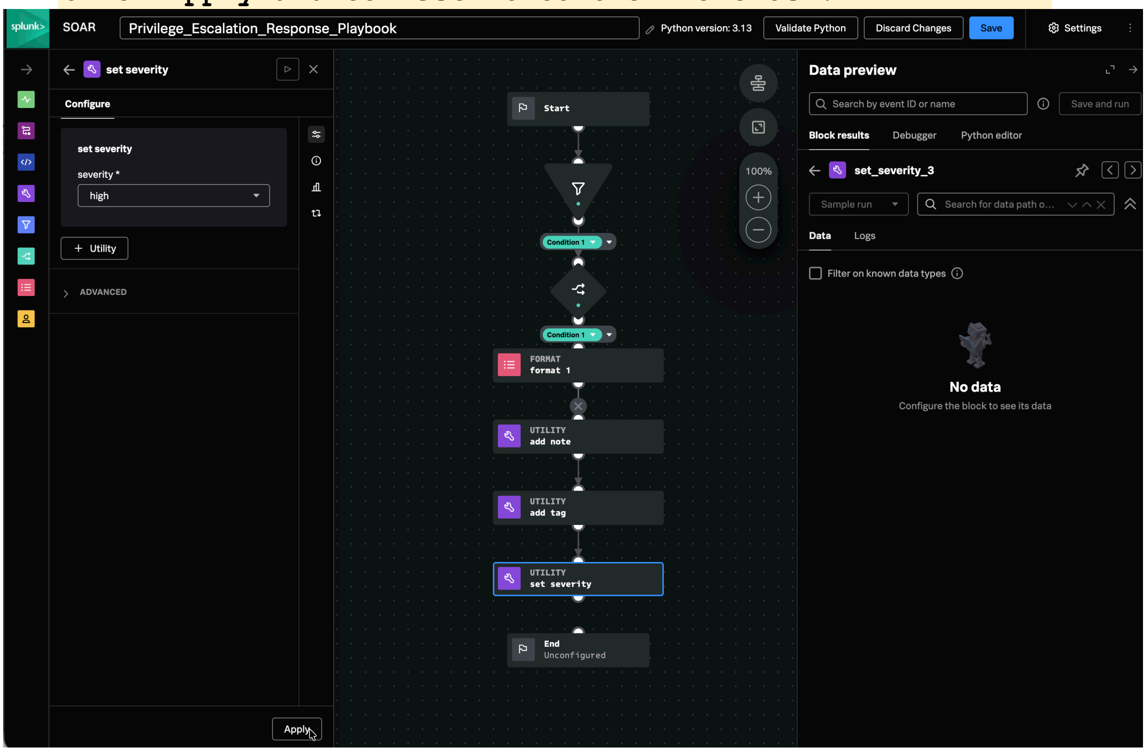
3) Add set severity Utility block

i) Drag and drop **Utility block** below the **add tag Utility block** and **Connect** them.

ii) Configure the Utility block
Go to `APIs tab > set severity`

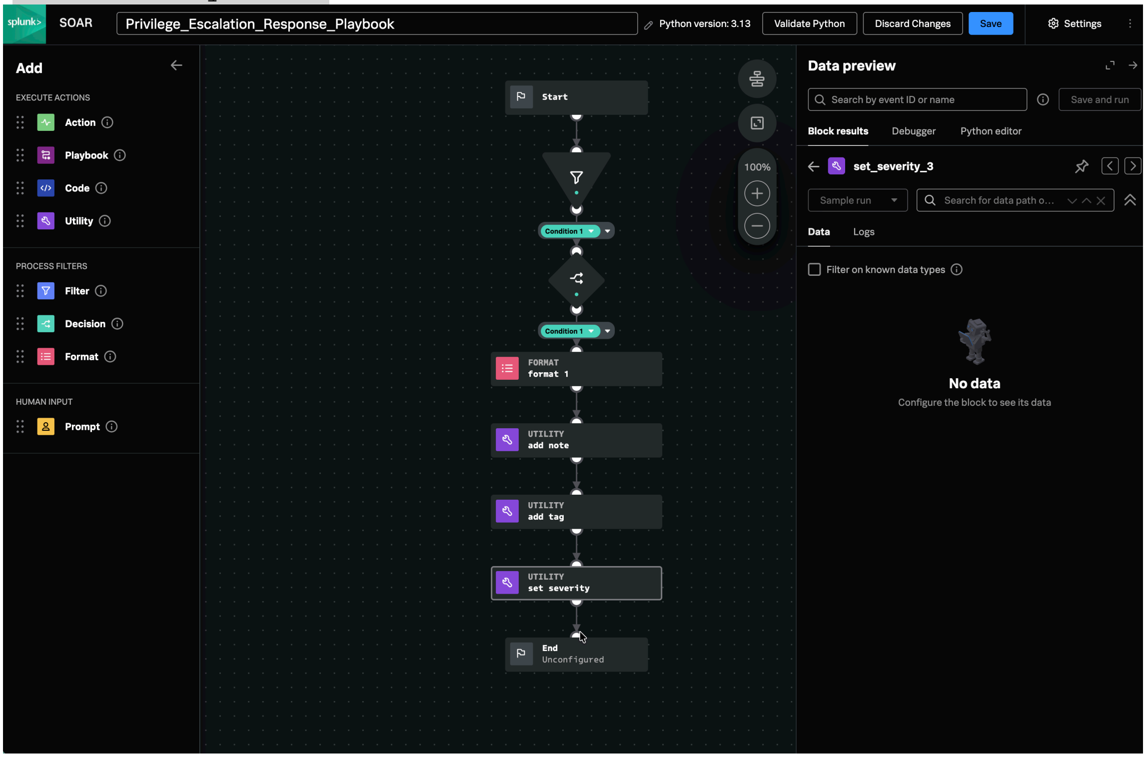


iii) Configure set severity
severity: high
Click **Apply** and **Connect** it to the **End block.**

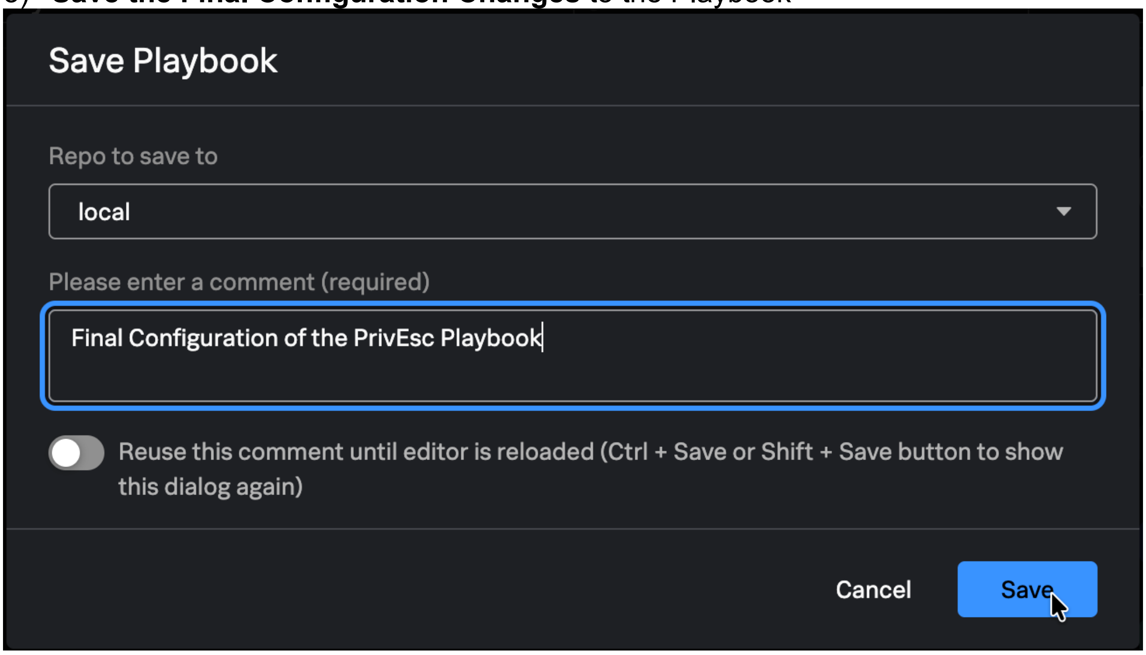


7. Ensure Connections between the Blocks in the Playbook and Commit the changes

1) Ensure a proper **playbook title** is given. If not make Changes.
2) Ensure the **Playbook blocks are connected** in this way:
`Start > Filter > Decision > Format > Add Note > Add Tag > Set Severity > End`



3) **Save the Final Configuration Changes** to the Playbook

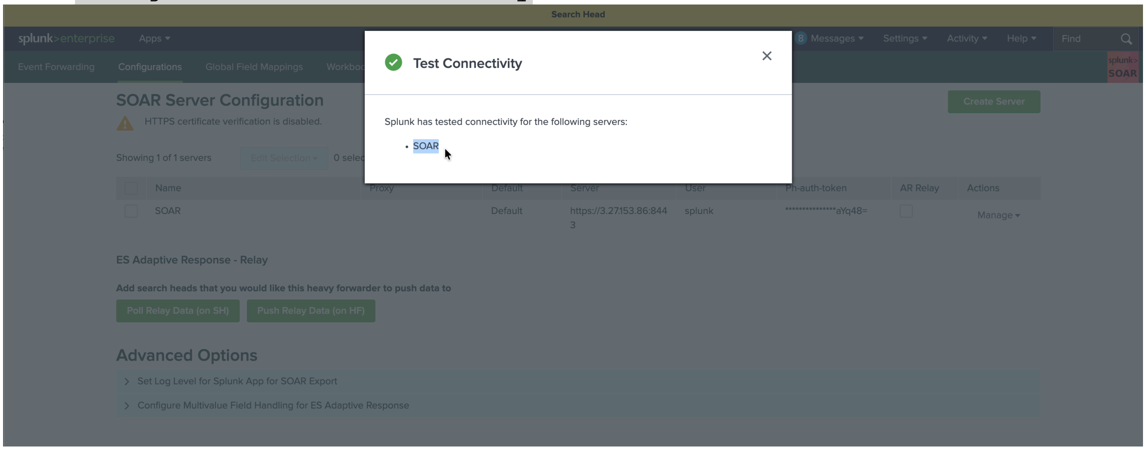


8. Test Playbook Execution

1) **Ensure the Splunk Enterprise (Search Head) is connected to the SOAR Instance**

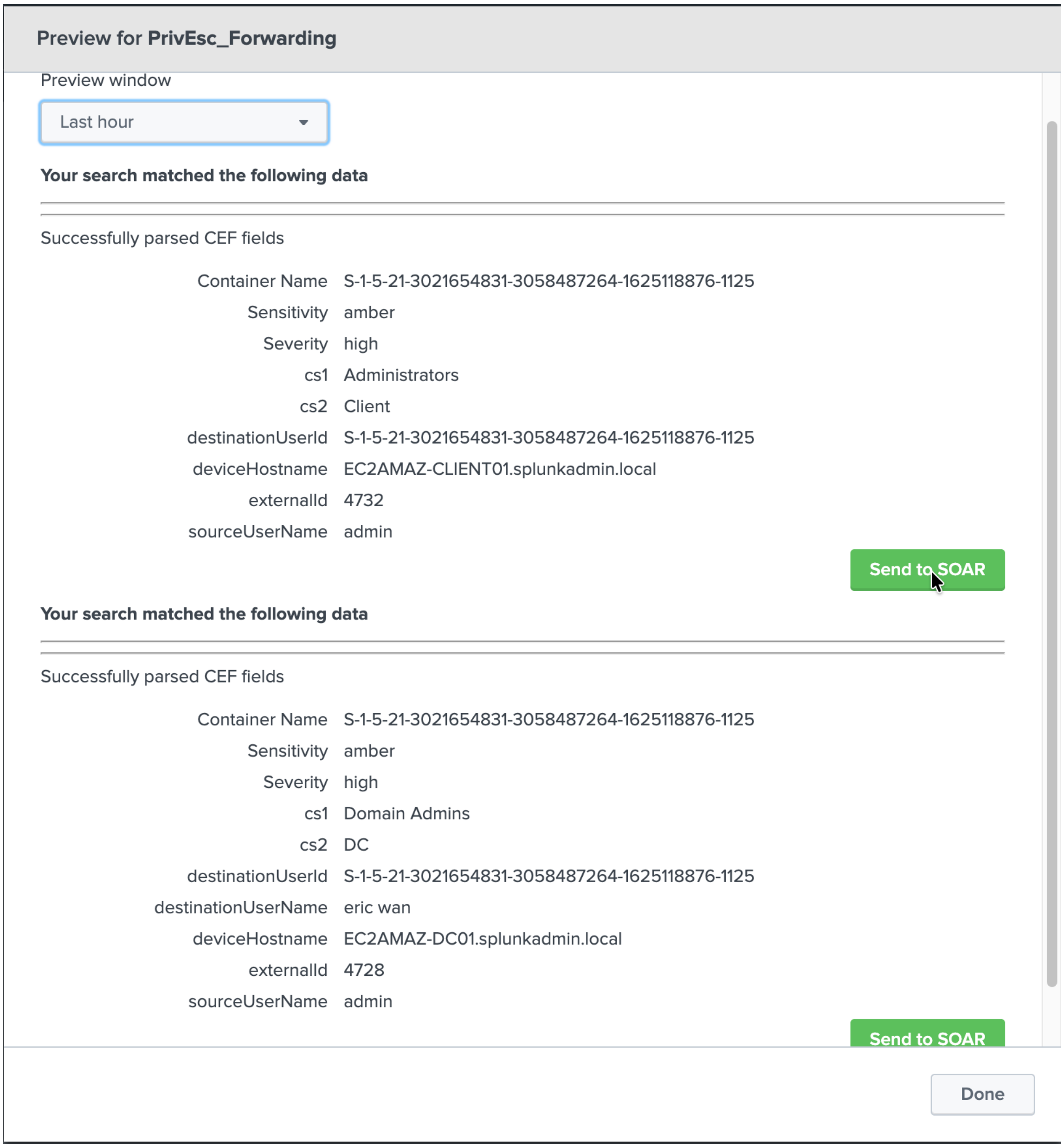
In **Search Head UI**,
Go to Apps > Splunk App for SOAR Export app > Configurations tab
Click on Manage > Edit and check the Server field.
Server: https://<SOAR_PublicIP>:8443 > Save

Click Manage > Test Connectivity



2) Ensure the playbook is executed by doing a Preview in Search Head Event Forwarding

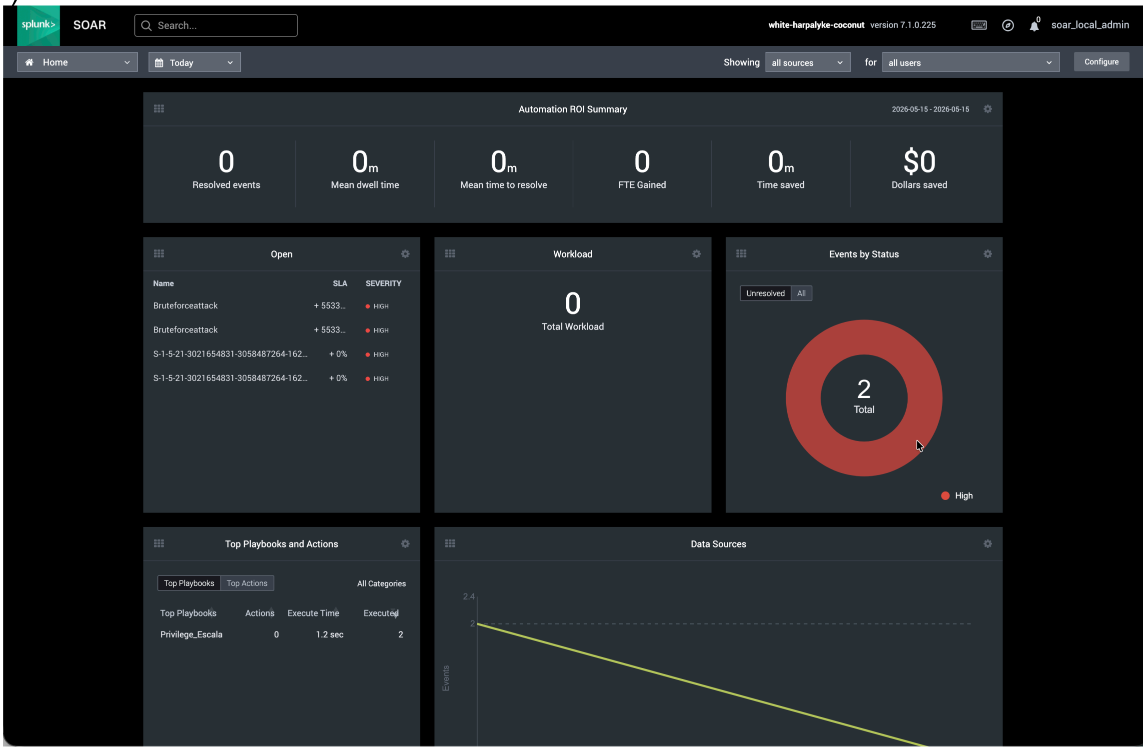
In **Event Forwarding tab**,
Click on the `Forwarder > Next > Save and Preview`
Specify a Time Range and Search
Click **Send to SOAR** on one of the Events



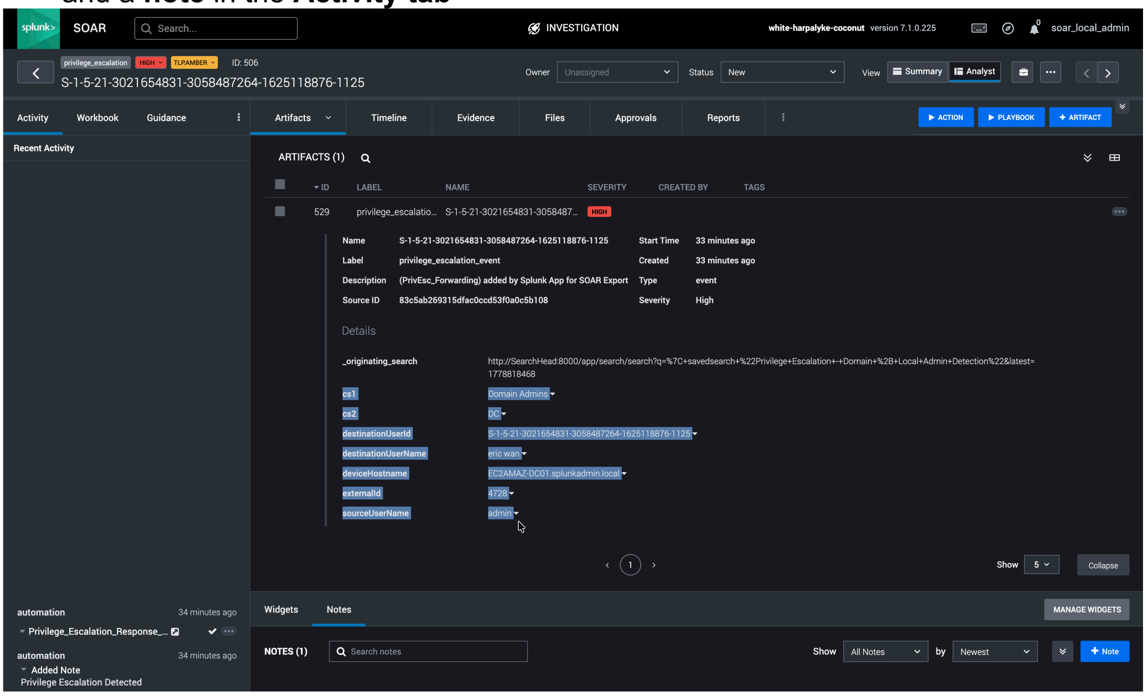
9. Ensure the Events are triggered by the Splunk alert via Event Forwarding in SOAR

In **SOAR UI,**

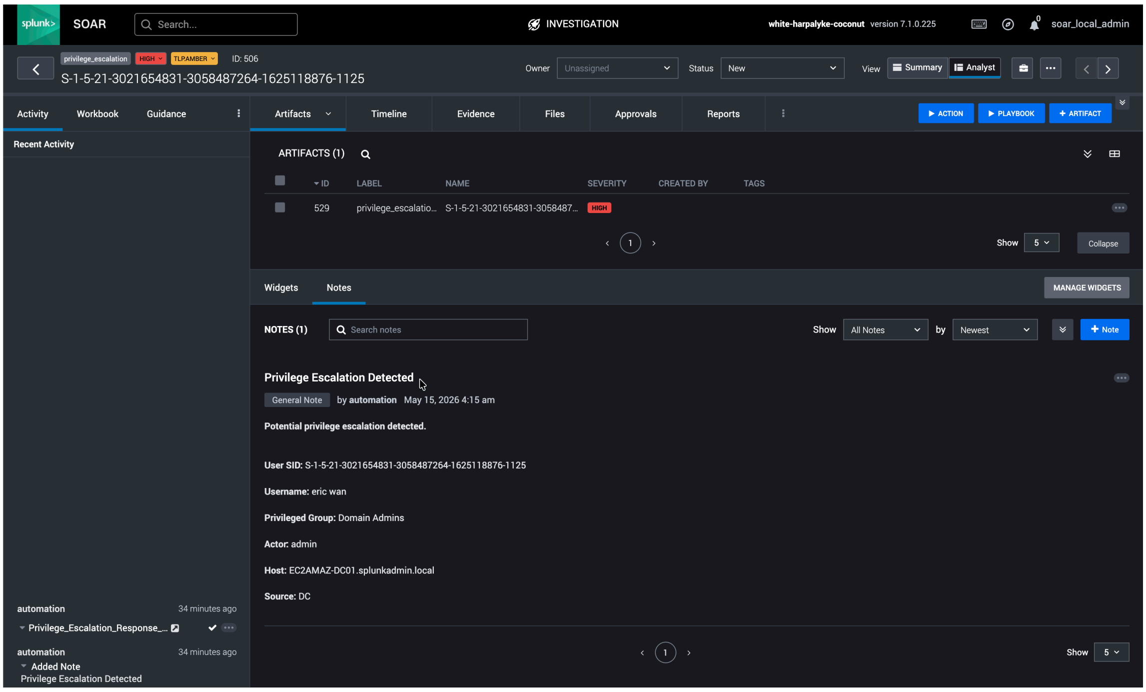
i) Should see **events** in the **Dashboard**. **Click on it**.



ii) In **Events page**, Click on a single event. Now You should see the **artifacts** and a **note** in the **Activity tab**



Notes tab



10. Simulate the PrivEsc attack and Test the Playbook Execution

- To see the `entire workflow from Privilege Escalation Attack Simulation > Alert triggering > Event Forwarding > SOAR Playbook Execution`, follow the steps referred in files - Refer files:
  - **23_Simulating_PrivEsc_Attack_WinServer_WinClient_V1.pdf** and
  - **24_Logs_Validation_using_Extraction_Detection_for_ServerClient_Events_V1.pdf**.

- For **Validating SOAR Playbook Execution part**, Refer **Step 9** above.

Next Steps

For the Test Results and Observations – Refer **28_Test_Results_Observations_and_UseCase_Summary_PrivilegeEscalation_V1.pdf**.